

Simple Reaction Time Task

PSY310: Lab in Psychology

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Github Link - <https://github.com/HetanshJalan/Psychology-Experiments>

Introduction

Reaction time is the time between a stimulus and a response. One of the stimuli of a simple reaction-time task elicits one of the responses; it is a procedure used to investigate the rate of perceptual and motor responding. Reaction-time measures have been used extensively in experimental psychology as behavioural indicators of processing speed (Deary et al., 2011).

A response to a visual stimulus involves more than simply seeing the stimulus. The participant has to identify a stimulus, interpret the information, formulate a motor response and perform the response. The recorded latency is thus not the time taken by one particular psychological process, but by several of these processes (Donders, 1969).

Reaction time is not necessarily consistent across trials. Differences in response latency might be due to attention, alertness, preparation for a response, knowledge of the task, as well as to properties of the experimental setup.

Taking several trials will provide a more accurate representation of performance than will a single observation.

Standardization of stimulus presentation and response recording is possible with computer mediated tasks. PsychoPy is an open source software package for behavioural experiments that offers tools for creating controlled stimuli and recording responses (Peirce et al., 2019).

Aim. Subjective simple visual reaction time was to be measured in one subject over five trials and the uniformity of the responses described.

Hypothesis. The participant was expected to respond in a matter of a few hundred milliseconds and there was an anticipation that this response would vary between the trials.

Method

Participant

The study was conducted with one 19 year old male undergraduate student who was recruited conveniently in a University setting. Participant 01 was identified as the participant. Explanation of task prior to data collection and informed consent. There were no personally identifying pieces of information in the dataset.

Materials

The and screen were used on a computer. The stimuli were created and reaction times recorded using PsychoPy software. An approximation of a 100×100 pixel cross-shaped visual target was used. To give a consistent starting point for each trial, a fixation cross was placed in front of the target.

Figure 1



Illustrative fixation screen

Figure 1. Illustrative reconstruction of the fixation screen.

Figure 2



Illustrative target screen

Figure 2. Illustrative reconstruction of the target screen.

Procedure

The participant was seated in front of the laptop in a relatively quiet setting, and was instructed to focus on the centre of the screen. The researcher told participant to respond as fast as they could when the target appeared.

A fixation cross was first shown for about 500ms prior to each trial. The stimulus was then presented and maintained until a response key was pressed. Reaction time was measured as the time between the presentation of the target and the response given on the keyboard. This sequence was repeated 5 times, without any trial feedback given other than standard verbal feedback.

Data treatment

The task yielded the reaction time (in msec) for every trial finished. 5 observations were kept for descriptive analysis. The mean, median, standard deviation, minimum, maximum and range were computed.

Results

The participant performed 5 trials. The range of reaction times was between 265 and 312 ms (47 ms). The mean reaction time was 291.80 ms (SD = 17.60), and the median was 294 ms.

Table 1

Trial	Reaction time (ms)
1	312
2	287
3	301
4	265
5	294

Table 1. Reaction-time observations across five trials.

Table 2

Measure	Value
Mean	

Median	294 ms
Standard deviation	
Minimum	265 ms
Maximum	312 ms
Range	47 ms

Table 2. Descriptive statistics for the five observations.

Figure 3

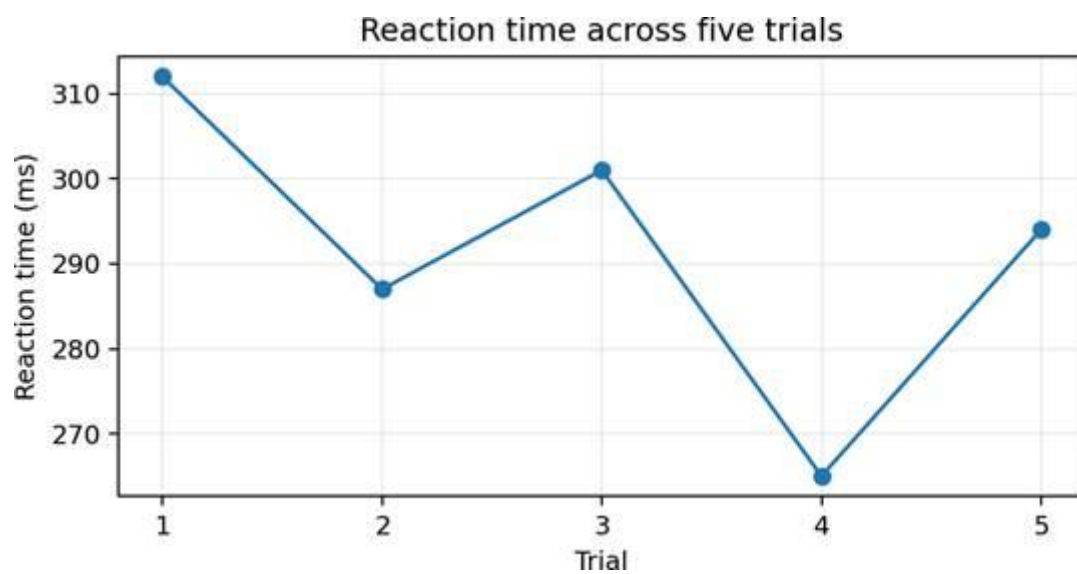


Figure 3. Reaction time across the five trials.

Discussion

Simple visual reaction time was measured within the experiment for one 19 year old person across five trials. The mean RT was 291.80ms, and the range of the reaction time was between

265 and 312ms. The hypothesis made predictions of responses within the few hundred milliseconds time frame and there was some trial-to-trial variability, and this pattern was consistent with the predictions of the hypothesis.

The relatively small range indicates a fairly constant performance level throughout the task. The difference in the fastest and slowest trials indicates, however, that the time required for responses was not the same across all trials. The fastest response was in Trial 4 and the slowest was in Trial 1.

Such differences may be due to differences in attention or response readiness, but this is not definitively established in this experiment.

The results are consistent with the general application of reaction time as a behavioural estimate of processing speed. Deary et al. (2011) state that simple reaction-time tasks involving the computer are practical methods that can be efficiently undertaken. The present demonstration was much smaller, but was basically the same in that a standardized stimulus was given and the latency of the response was recorded.

PsychoPy proved to be very helpful as it enabled stimuli to be presented and responses recorded with computer timing instead of manual timing. PsychoPy has been designed specifically for use in presenting stimuli and collecting behavioural responses (Peirce et al., 2019) for behavioural experiments.

The limitations and future directions are discussed.

The major drawback of the study was the limited number of one participant in the study. The results are therefore only applicable for a single performance and cannot be extrapolated to university students as a group. Additionally, only five trials are used, which limits the estimate of the typical reaction time. The more trials, the more reliable is the estimate, and the more participants in the sample size, the more meaningful the comparisons would be.

Future studies could consider a comparison of visual reaction time to auditory reaction time, a comparison of focused and distracted conditions, or a comparison of simple reaction time to a choice reaction time task.

More trials would also allow for a more detailed examination of within-person variability.

The experiment as a whole is illustrative of the achievement of a quantitative measure of simple visual reaction time after a short, computer-based task. On all 5 trials the participant's response time ranged within a couple hundred milliseconds with slight variation among the

observations. While not universalistic, it gives a good example of how to gather and analyse behavioural reaction time data.

References

Deary, I. J., Liewald, D., & Nissan, J. (2011). A free, easy-to-use, computer-based simple and four-choice reaction time programme: The Deary-Liewald reaction time task. *Behavior research methods*, 43(1), 258-268.

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